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INCREASING ULTRASONIC FLOW MEASUREMENT UPDATE RATE

Abstract

A method includes at a first time, transmitting a first set of ultrasonic pulses from a first location. The method also includes, at a second time, receiving the first set of ultrasonic pulses at a second location. The method also includes, at a third time, transmitting a second set of ultrasonic pulses from the second location. The method also includes, at a fourth time, receiving a first echo of multiple echoes of the first set of ultrasonic pulses at the second location, in which the third time is the same as the first time, or is between the second time and the fourth time. The method also includes, at a fifth time, receiving the second set of ultrasonic pulses at the first location; and providing a flow rate measurement between the first and second locations based on the first time, the second time, the third time, and the fifth time.

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Background/Summary

BACKGROUND

[0001] Ultrasonic flow sensing based on measurements of ultrasonic pulses can be used in a variety of applications including industrial, medical, and residential applications. Some examples of ultrasonic flow sensing use two transducers to measure upstream and downstream time-of-flight (TOF) information of ultrasonic pulses transmitted and propagated through a flow in a tube, and then calculates flow rate based on the TOF information. Existing ultrasonic flow sensing solutions often have a low update rate, e.g., less than 50 Hz, due to multiple reflections/echoes following each ultrasonic pulse transmission in the tube. It takes time for the echoes to die down before the next transmission of ultrasonic pulses can be initiated and transmitted. However, ultrasonic flow sensing in a lot of applications, such as in continuous positive airway pressure (CPAP) machines or ventilator, often requires a much higher update rate, e.g., greater or equal to 2 KHz.

SUMMARY

[0002] In an example, an apparatus includes a first ultrasonic transmitter at a first location, a first ultrasonic receiver at the first location, a second ultrasonic transmitter at a second location, a second ultrasonic receiver at the second location, and one or more controllers in communication with the first ultrasonic transmitter, the first ultrasonic receiver, the second ultrasonic transmitter, and the second ultrasonic receiver. The one or more controllers are configured to, at a first time, transmit a first set of ultrasonic pulses using the first ultrasonic transmitter. The one or more controllers are further configured to, at a second time, receive the first set of ultrasonic pulses using the second ultrasonic receiver. The one or more controllers are further configured to, at a third time, transmit a second set of ultrasonic pulses using the second ultrasonic transmitter. The one or more controllers are further configured to, at a fourth time, receive a first echo of multiple echoes of the first set of ultrasonic pulses using the second ultrasonic receiver, in which the third time is the same as the first time, or is between the second time and the fourth time. The one or more controllers are further configured to, at a fifth time, receive the second set of ultrasonic pulses using the first ultrasonic receiver; and provide a flow rate measurement between the first and second locations based on the first time, the second time, the third time, and the fifth time.

[0003] In another example, an apparatus includes a first ultrasonic transmitter, a first ultrasonic receiver; and a controller coupled to the first ultrasonic transmitter and the first ultrasonic receiver, the controller having a controller input and a measurement output. The controller is configured to receive a first signal at the controller input, the first signal representing a first time of transmission of a first set of ultrasonic pulses by a second ultrasonic transmitter. The controller is further configured to, at a second time, receive the first set of ultrasonic pulses using the first ultrasonic receiver. The controller is configured to, based on the first time and the second time, determine an expected arrival time of a first echo of multiple echoes of the first set of ultrasonic pulses at the first ultrasonic receiver. The controller is further configured to determine a third time of transmission of a second set of ultrasonic pulses, in which the third time precedes the expected arrival time, and at the third time, transmit a second set of ultrasonic pulses using the first ultrasonic transmitter. The controller is further configured to provide a second signal representing a flow rate measurement at the measurement output based on the first, second, and third times.

[0004] In another example, a method includes, at a first time, transmitting a first set of ultrasonic pulses from a first location. The method also includes, at a second time, receiving the first set of ultrasonic pulses at a second location. The method also includes, at a third time, transmitting a second set of ultrasonic pulses from the second location. The method also includes, at a fourth time, receiving a first echo of multiple echoes of the first set of ultrasonic pulses at the second location, in which the third time is the same as the first time, or is between the second time and the

fourth time. The method also includes, at a fifth time, receiving the second set of ultrasonic pulses at the first location; and providing a flow rate measurement between the first and second locations based on the first time, the second time, the third time, and the fifth time.

[0005] In another example, a non-transitory computer readable medium stores instructions that, when executed by processor circuitry, cause the processor circuitry to control a first ultrasonic transmitter and a first ultrasonic receiver at a first location, a second ultrasonic transmitter and a second ultrasonic receiver at a second location to: at a first time, transmit a first set of ultrasonic pulses from a first location, at a second time, receive the first set of ultrasonic pulses at a second location, at a third time, transmit a second set of ultrasonic pulses from the second location, at a fourth time, receive a first echo of multiple echoes of the first set of ultrasonic pulses at the second location, in which the third time is the same as the first time, or is between the second time and the fourth time, at a fifth time, receive the second set of ultrasonic pulses at the first location; and provide a flow rate measurement between the first and second locations based on the first time, the second time, the third time, and the fifth time.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a schematic diagram of an ultrasonic flow measuring system, in an example.

[0007] FIG. 2 is a schematic diagram of deployment of the ultrasonic flow measuring system of FIG. 1 for flow measurement, in an example.

[0008] FIG. 3 is a schematic diagram of an integrated ultrasonic flow measuring unit, in an example.

[0009] FIG. 4 is a schematic diagram of the controller in FIGS. 1 and 3, in an example.

[0010] FIG. 5A is a graph illustrating timing sequences of two sets of ultrasonic pulses transmitted and received by the transmitters and receivers of FIGS. 1 and 3 during one flow measurement cycle, in an example.

[0011] FIG. 5B is a graph illustrating timing sequences of additional sets of ultrasonic pulses transmitted and received by the transmitters and receivers of FIGS. 1 and 3 during multiple flow measurement cycles, in an example.

[0012] FIG. 6 is a graph illustrating an example of multiple resonance peaks in frequency responses of ultrasonic transmitters and receivers of FIGS. 1 and 3, in an example.

[0013] FIG. 7 is a flowchart of a method for ultrasonic flow measuring by the system of FIGS. 1 and 3, in an example.

[0014] FIG. 8 is a schematic of an example processor that can implement part of the system of FIGS. 1 and 3, in an example.

DETAILED DESCRIPTION

[0015] The same reference numbers or other reference designators are used in the drawings to designate the same or similar (either by function and/or structure) features.

[0016] FIG. 1 is a schematic diagram of an example of an illustrative ultrasonic flow measuring system **100**, which includes a first ultrasonic transmitter **102** and a first ultrasonic receiver **104**, both at a first location. In one example, the first ultrasonic transmitter **102** and the first ultrasonic receiver **104** can be part of a single transducer, which can be controlled to transmit ultrasonic signals and to receive ultrasonic signals. In another example, the first ultrasonic transmitter **102** and the first ultrasonic receiver **104** can be separate transducers. The system **100** further includes a second ultrasonic transmitter **106** and a second ultrasonic receiver **108**, both at a second location. The second ultrasonic transmitter **106** and the second ultrasonic receiver **108** can be part of a single transducer or can be separate transducers. In one example, each of the ultrasonic transmitters includes a driver having an output for transmitting a set of ultrasonic pulses and each of the

ultrasonic receivers includes an amplifier having an input for detecting and receiving a set of ultrasonic pulses. In one example, a single transducer is configured to have both a transmitter output and a receiver input.

[0017] The system **100** further includes one or more controllers **110** in communication with the first ultrasonic transmitter **102**, the first ultrasonic receiver **104**, the second ultrasonic transmitter **106**, and the second ultrasonic receiver **108**. Such communication can be either wired communication via physical connections or wireless communication over one or more communication networks (not shown). For example, each of the communication networks can be an Internet, an intranet, a wide area network (WAN), a local area network (LAN), Bluetooth, Wi-Fi, and a mobile communication network. In one example, for TX, each ultrasound transmitter may be driven by a transceiver at configurable voltages to transmit signals to the controllers **110**; for RX, signals from the controllers **110** may go through analog frontend circuitry of each ultrasound receiver for amplification and may then be digitalized with an analog-digital converter (ADC). In one example, the one or more controllers **110** include a single controller.

[0018] FIG. **2** is a schematic diagram of an example of deployment of the ultrasonic flow measuring system of FIG. **1** for flow measurement through a tube. In the example of FIG. **2**, the ultrasonic flow measuring system uses a pair of two transducers **202** and **204** to measure upstream and downstream TOF information of ultrasonic pulses through a flow in a tube and then calculates flow rate based on the TOF information. In one example, the first transducer **202** and the second transducer **204** are located at a first location A and a second location B of the tube, respectively. As shown in FIG. **2**, the distance between the first location A and the second location B is denoted as L. The diameter of the tube is denoted as D. In one example, the angle of propagation between the direction of the flow and the direction of which ultrasonic pulses are being transmitted and received by the two transducers **202** and **204**, is denoted as θ .

[0019] When the first transducer **202** at the first location A transmits (or fires) a set of ultrasonic pulses downstream through the flow towards the second transducer **204** at the second location B, the downstream TOF $T_{sub.AB}$ can be calculated as:

$$[00001] T_{AB} = \frac{L}{c + v \cos(\theta)}$$

wherein c is the speed of sound in the flow media and v is the speed of the flow through the tube (also referred to as the flow speed). Conversely, when the second transducer **204** at the second location B transmits (or fires) a set of ultrasonic pulses upstream through the flow towards the first transducer **202** at the first location A, the upstream TOF $T_{sub.BA}$ can be calculated as:

$$[00002] T_{BA} = \frac{L}{c - v \cos(\theta)}$$

The flow speed v can be calculated based on $T_{sub.AB}$ and $T_{sub.BA}$ as:

$$[00003] v = \frac{L}{2 \cos(\theta)} \left(\frac{T_{BA} - T_{AB}}{T_{BA} T_{AB}} \right)$$

Flow rate can then be calculated as $v \cdot A$ —product of the flow speed v and area A of the cross section of the tube.

[0020] In one example, $T_{sub.AB}$ and $T_{sub.BA}$ can be found through absolute TOF measurement, for example, by determining when envelope/amplitude of the ultrasonic pulses received at one of the transducers crosses a pre-set threshold. For differential TOF, $T_{sub.AB} - T_{sub.BA}$ can be found through cross correlating upstream echoes and downstream echoes of the ultrasonic pulses, wherein signal of the received echo from one transducer is passed to the other transducer for cross correlation. In one example, the received analog signal from one transducer is passed to the other transducer having signal processing capabilities, which then performs post processing on the analog signal using an analog-digital converter (ADC). In one example, the processing of the data, e.g., calculation of TOF $T_{sub.AB}$, calculation of TOF $T_{sub.BA}$, calculation of flow rate, etc., may be performed by the one or more controller **110**.

[0021] FIG. **3** is a schematic diagram of an example of an integrated ultrasonic flow measuring unit **300**, which includes an ultrasonic transmitter **302**, an ultrasonic transmitter **304**, and a controller

306. In an example, the controller **306** is configured to accept as its input—controller input—from other units, calculate the flow rate based on the controller input received, and provide flow measurement result as measurement output to other units. In one example, the ultrasonic transmitter **302**, the ultrasonic transmitter **304**, and the controller **306** are all included in a single chip.

[0022] FIG. 4 is a schematic diagram of an example of the controller **110** in FIG. 1 or **306** in FIG. 3, wherein the controller includes one or more of driver **402**, signal processor **404**, transmit timing control module **406**, time-of-flight determination module **408**, flow rate determination module **410**, and communication interface **412**. As shown in the example of FIG. 4, controller **110** or **306** is configured to receive, as part of its controller input, one or more of time of transmission, TOF information, position information (e.g., L or θ), flow measurement, and other programming data from one or more other transducers through communication interface **412**. Driver **402** is configured to interact with ultrasonic transmitter and ultrasonic receiver to cause the ultrasonic transducers to transmit and detect/receive sets of ultrasonic pulses/signals, respectively. In one example, driver **402** may also condition the signals via, e.g., signal biasing or pre-biasing. For signal processing, the signal processor **404** is configured to distinguish echoes from a direct path signal of the ultrasonic pulses, e.g., based on expected time of arrival of the signals. For example, the signal processor **404** receives and processes echoes of the first set of ultrasonic pulses that are transmitted by the first ultrasonic transmitter **102**, bounced back from the second ultrasonic receiver **108**, and received at the first ultrasonic receiver **104** to estimate the time of arrival of the signals. The signal processor **404** dismisses the echoes from other measurement cycles or sessions, e.g., based on frequency or band of frequencies, as discussed below, and determines a time of reception of a direct path signal from the other transducer(s). In one example, transmit timing control module **406** determines and controls timing of the transmission of the set of ultrasonic pulses via the ultrasonic transmitter **302**. Time-of-flight determination module **408** determines TOF based on the transmit and receive time of the set of ultrasonic pulses and flow rate determination module **410** determines the flow rate based on the TOF information received from the time-of-flight determination module **408**. In one example, controller **110** or **306** is configured to provide/transmit the flow rate measurement as part of the measurement output through the communication interface **412**.

[0023] FIG. 5A is a graph illustrating timing sequences of two sets of ultrasonic pulses transmitted and received by the transmitters and receivers of FIGS. 1 and 3 during one flow measurement cycle (the discussions below based on FIG. 1 is also applicable to FIG. 3). At a first time t_1 , the one or more controllers **110** are configured to transmit a first set of ultrasonic pulses using the first ultrasonic transmitter **102**. The first set of ultrasonic pulses is received by the controllers **110** using the second ultrasonic receiver **108** at a second time t_2 . The one or more controllers **110** are further configured to transmit a second set of ultrasonic pulses using the second ultrasonic transmitter **106** at a third time t_3 , which is a time interval ΔT after the first set of ultrasonic pulses is transmitted at the first time t_1 . For example, the second set of ultrasonic pulses can be transmitted around ΔT of 230 us from the start of transmitting of the set of ultrasonic pulses at the first time t_1 .

[0024] Note that there may be multiple reflections of the first set of ultrasonic pulses bouncing/echoing back and forth between the first ultrasonic receiver **104** at location A and the second ultrasonic receiver **108** at location B and it takes time for the echoes to die down, e.g., amplitude of the echoes received at one of the transducers to be attenuated below a pre-set noise threshold. For example, a first echo of multiple echoes of the first set of ultrasonic pulses is received by the one or more controllers **110** using the second ultrasonic receiver **108** at location B at a fourth time t_4 . As shown by the example of FIG. 5A, the second set of ultrasonic pulses is transmitted by the one or more controllers **110** at the third time t_3 , which is between the second time t_2 and the fourth time t_4 . In other words, instead of waiting for all echoes of the first set of ultrasonic pulses to die down before firing new set of ultrasonic pulses, the one or more controllers **110** are configured to transmit the second set of ultrasonic pulses using the second ultrasonic

transmitter **106** before the first echo of the first set of ultrasonic pulses is received using the second ultrasonic receiver **108** at fourth time **t4**. The second set of ultrasonic pulses is then received by the one or more controllers **110** at a fifth time **t5** using the first ultrasonic receiver **104**.

[0025] In one example, the one or more controllers **110** are configured to transmit the second set of ultrasonic pulses at the third time **t3** (AT from the first time **t1**) after the first set of ultrasonic pulses finishes a round trip and bounces back to location A. Delaying the transmission of the second set of ultrasonic pulses by a round-trip time ensures that the echoes of the first set of ultrasonic pulses have become a lot smaller and thus will not interfere with the second set of ultrasonic pulses.

[0026] In one example, the first set of ultrasonic pulses is transmitted downstream of a flow from location A towards location B and the second set of ultrasonic pulses is transmitted upstream from location B towards A during one flow measuring cycle. The one or more controllers **110** are configured to determine both the downstream TOF $T_{sub.AB}$ and the upstream TOF $T_{sub.BA}$ based on one or more of the first time **t1**, the second time **t2**, the third time **t3**, and the fifth time **t5**. The one or more controllers **110** then provide a flow rate measurement based on the downstream TOF $T_{sub.AB}$ and the upstream TOF $T_{sub.BA}$ as discussed above.

[0027] In one example, the one or more controllers **110** receive the first time **t1** of transmission of the first set of ultrasonic pulses as part of its controller input and determine TOF of the first set of ultrasonic pulses based on **t1** and the second time **t2** when the first set of ultrasonic pulses is received by the second ultrasonic receiver **108**. In one example, the one or more controllers **110** further use the TOF as well as other programming data received as part of controller input to estimate the fourth time **t4**—the expected time the first echo of the first set of ultrasonic pulses is to be received by the second ultrasonic receiver **108**. The one or more controllers **110** then transmit the second set of ultrasonic pulses at the third time **t3** before the expected time **t4**.

[0028] By time-interleaving initiation and transmission of ultrasonic pulses instead of waiting for echoes from the prior ultrasonic pulses to die down, the update rate of ultrasonic flow measurement can be increased while reducing interferences between echoes of upstream/downstream ultrasonic pulses. Such an approach is especially suited for flow sensing in applications with high update rate, e.g., greater or equal to 2 KHz, such as in CPAP machines and ventilators.

[0029] In one example, the one or more controllers **110** are configured to initiate another flow measurement cycle by transmitting/firing additional sets of ultrasonic pulses before the echoes from the first and second sets of ultrasonic pulses die down. FIG. 5B is a graph illustrating timing sequences of additional sets of ultrasonic pulses transmitted and received by the transmitters and receivers of FIGS. 1 and 3 during multiple flow measurement cycles. At a sixth time **t6**, the one or more controllers **110** are configured to transmit a third set of ultrasonic pulses using the first ultrasonic transmitter **102**. The third set of ultrasonic pulses is received by the controllers **110** using the second ultrasonic receiver **108** at a seventh time **t7**. The one or more controllers **110** are further configured to transmit a fourth set of ultrasonic pulses using the second ultrasonic transmitter **106** at an eighth time **t8**. As shown by the example of FIG. 5B, the eighth time **t8** is between the seventh time **t7** and the ninth time **t9**, when a first echo of multiple echoes of the third set of ultrasonic pulses is received by the one or more controllers **110** using the second ultrasonic receiver **108** at a ninth time **t9**. In other words, instead of waiting for all echoes of the third set of ultrasonic pulses to die down before firing new set of ultrasonic pulses, the one or more controllers **110** are configured to transmit the fourth set of ultrasonic pulses using the second ultrasonic transmitter **106** before the first echo of the third set of ultrasonic pulses is received using the second ultrasonic receiver **108** at ninth time **t9**. The fourth set of ultrasonic pulses is then received by the one or more controllers **110** at a tenth time **t10** using the first ultrasonic receiver **104**.

[0030] In one example, the third set of ultrasonic pulses is transmitted downstream of the flow from location A towards location B similar to the first set of ultrasonic pulses and the fourth set of ultrasonic pulses is transmitted upstream from location B towards A similar to the second set of ultrasonic pulses during the current flow measuring cycle. The one or more controllers **110** are

configured to determine both the downstream TOF T.sub.AB and the upstream TOF T.sub.BA based on one or more of the sixth time **t6**, the seventh time **t7**, the eighth time **t8**, and the tenth time **t10**, and to update the flow rate measurement based on the downstream TOF T.sub.AB and the upstream TOF T.sub.BA during the current flow measurement cycle as discussed above.

[0031] In some examples, the one or more controllers **110** can control the pair of two transducers **202** and **204** to transmit, respectively, the first set of ultrasonic pulses and the second set of ultrasonic pulses of FIG. 5A simultaneously, which can further reduce the update rate of ultrasonic flow measurement compared with the interleaving arrangement illustrated in FIG. 5A. Such arrangements can be supported by having two separate receive signal chains (e.g., separate analog front ends, separate ADCs, etc.) coupled to each of transducers **202** and **204**. Interference between the transmitted and received ultrasonic pulses can be reduced by, for example, introducing a blanking period at each receive signal chain after transmission of a set of ultrasonic pulses, to avoid the receive signal chain of a transducer processing the locally transmitted ultrasonic pulses as signals received from the other transducer.

[0032] The one or more controllers **110** are configured to reduce interference between the multiple sets of ultrasonic pulses transmitted by the same ultrasonic transmitter in the same direction through signal modulation. In one example, the one or more controllers **110** are configured to set and transmit the first set of ultrasound pulses and the second set of ultrasound pulses at a first pulse frequency, while setting and transmitting the third set of ultrasound pulses and the fourth set of ultrasound pulses at a second pulse frequency via the first and second ultrasonic transmitters. In one example, the first pulse frequency and the second pulse frequency are different, e.g., the first pulse frequency is 290 kHz while the second pulse frequency is at a different frequency 310 kHz. Note that both the first and the second pulse frequencies are within the bandwidth of the first and second ultrasonic transmitters.

[0033] Each of the ultrasonic transducers (transmitters and receivers) may have multiple resonance peaks. The one or more controllers **110** are configured to select the first and the second pulse frequencies based on the ultrasonic transducer's characteristic to obtain better performance and accuracy. Specifically, the one or more controllers **110** are configured to select the first pulse frequency and the second pulse frequency based on frequency responses of one or more of the ultrasonic transducers, e.g., the first ultrasonic transmitter, the first ultrasonic receiver, the second ultrasonic transmitter, and the second ultrasonic receiver. FIG. 6 is a graph illustrating an example of multiple resonance peaks in frequency responses of the ultrasonic transmitters and receivers of FIGS. 1 and 3. As shown by the example of FIG. 6, the one or more controllers **110** may sweep a frequency range from 285 kHz to 300 kHz and automatically select 289 kHz (**602**) as the first pulse frequency because the received signal is strongest at **602**. The one or more controllers may continue to sweep a range from 301 kHz to 310 kHz and select 306 kHz (**604**) as the second pulse frequency. Such frequency selection leads to a better signal to noise ratio, which translates to a better TOF measurement accuracy and flow measurement accuracy.

[0034] In one example, the one or more controllers **110** are configured to set and transmit the first set of ultrasound pulses and the second set of ultrasound pulses as upchirp pulse signals having a first pulse frequency range. The one or more controllers **110** are further configured to set and transmit the third set of ultrasound pulses and the fourth set of ultrasound pulses as downchirp pulse signals having a second pulse frequency range. In one example, the first pulse frequency range and the second pulse frequency range overlap, e.g., with the first and the second set of ultrasound pulses using upchirp from 290 kHz to 310 kHz and the third and the fourth set of ultrasonic pulses using downchirp from 310 kHz to 290 kHz. In one example, the first pulse frequency range and the second pulse frequency range do not overlap, e.g., with the first and the second set of ultrasound pulses using upchirp from 290 kHz to 300 kHz and the third and the fourth set of ultrasonic pulses using downchirp from 310 kHz to 300 kHz. The frequency separation coupled with signal modulation between the sets of ultrasonic pulses creates isolation of signals

and residual echoes during calculation of cross-correlation between upstream/downstream signals. [0035] FIG. 7 is a flowchart illustrating a method **700** for ultrasonic flow measurement. Method **700** can include example methods and operations described in, for example, FIG. 1. Method **700** can be performed by various components of system **100** including, for example, the first ultrasonic transmitter **102**, the first ultrasonic receiver **104**, the second ultrasonic transmitter **106**, the second ultrasonic receiver **108**, and the one or more controllers **110**.

[0036] At block **702**, the one or more controllers **110** are configured to, at a first time, transmit a first set of ultrasonic pulses using the first ultrasonic transmitter **102** from the first location.

[0037] At block **704**, the one or more controllers **110** are configured to, at a second time, receive the first set of ultrasonic pulses transmitted by the first ultrasonic transmitter **102** using the second ultrasonic receiver **108** at the second location.

[0038] At block **706**, the one or more controllers **110** are configured to, at a third time, transmit a second set of ultrasonic pulses using the second ultrasonic transmitter **106** at the second location.

[0039] At block **708**, the one or more controllers **110** are configured to, at a fourth time, receive a first echo of multiple echoes of the first set of ultrasonic pulses using the second ultrasonic receiver **108** at the second location, in which the third time is the same as the first time, or is between the second time and the fourth time.

[0040] At block **710**, the one or more controllers **110** are configured to, at a fifth time, receive the second set of ultrasonic pulses transmitted by the second ultrasonic transmitter **106** using the first ultrasonic receiver **104** at the first location.

[0041] At block **712**, the one or more controllers **110** are configured to provide a flow rate measurement between the first and second locations based on the first time, the second time, the third time, and the fifth time.

[0042] FIG. 8 is a block diagram of an example processor platform **800** including processor circuitry structured to execute machine-readable instructions to implement the logic depicted in the examples of FIGS. 1 and 3. Processor platform **800** of the illustrated example can include processor circuitry **812**. The processor circuitry **812** of the illustrated example includes hardware. For example, processor circuitry **812** can be implemented by one or more integrated circuits, logic circuits, FPGAs, microprocessors, Central Processing Units (CPUs), Graphical Processing Units (GPUs), Digital Signal Processors (DSPs), and/or microcontrollers from any desired family or manufacturer. Processor circuitry **812** can be implemented by one or more semiconductor-based (e.g., silicon-based) devices. In this example, processor circuitry **812** can implement the controllers **110** of FIG. 1 and controller **306** of FIG. 3.

[0043] Processor circuitry **812** of the illustrated example can include a local memory **813** (e.g., a cache, registers, etc.). Processor circuitry **812** of the illustrated example is in communication with a computer-readable storage device such as a main memory including a volatile memory **814** and a non-volatile memory **816** by a bus **818**. The volatile memory **814** can be implemented by, for example, Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®), and/or any other type of RAM device. The non-volatile memory **816** may be implemented by programmable read-only memory, flash memory and/or any other desired type of non-volatile memory device. Access to the main memory **814**, **816** of the illustrated example can be controlled by a memory controller **817**.

[0044] The processor platform **800** of the illustrated example also includes interface circuitry **820**. The interface circuitry **820** may be implemented by hardware in accordance with any type of interface standard, such as an Inter-Integrated Circuit (I2C) interface, a Serial Peripheral Interface (SPI), an Ethernet interface, a universal serial bus (USB) interface, a Bluetooth® interface, a near field communication (NFC) interface, a Peripheral Component Interconnect (PCI) interface, and/or a Peripheral Component Interconnect Express (PCIe) interface.

[0045] In the illustrated example, one or more input ADCs **822** are connected to bus **818**. The

ADCs **822** can convert analog signals to digital signals for processing by the processor circuitry **812**.

[0046] One or more output devices **824** can be connected to the interface circuitry **820** of the illustrated example. The output device(s) **824** can include circuits such as driver circuits for, e.g., the driver **402** in FIG. 4.

[0047] Machine-readable instructions **832** can be stored in volatile memory **814** and/or non-volatile memory **816**. Upon execution by the processor circuitry **812**, the machine-readable instructions **832** cause the processor platform **800** to perform any or all of the functionality described herein attributed to the system **100** and unit **300**.

[0048] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0049] Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0050] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0051] In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0052] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. An apparatus, comprising: a first ultrasonic transmitter at a first location; a first ultrasonic receiver at the first location; a second ultrasonic transmitter at a second location; a second ultrasonic receiver at the second location; and one or more controllers in communication with the first ultrasonic transmitter, the first ultrasonic receiver, the second ultrasonic transmitter, and the second ultrasonic receiver and configured to: at a first time, transmit a first set of ultrasonic pulses using the first ultrasonic transmitter; at a second time, receive the first set of ultrasonic pulses using the second ultrasonic receiver; at a third time, transmit a second set of ultrasonic pulses using the second ultrasonic transmitter; at a fourth time, receive a first echo of multiple echoes of the first set of ultrasonic pulses using the second ultrasonic receiver, in which the third time is the same as the first time or is between the second time and the fourth time; at a fifth time, receive the second set of ultrasonic pulses using the first ultrasonic receiver; and provide a flow rate measurement between the first and second locations based on the first time, the second time, the third time, and the fifth time.
2. The apparatus of claim 1, wherein the one or more controllers include a single controller.
3. The apparatus of claim 1, wherein the one or more controllers are configured to: at a sixth time, transmit a third set of ultrasonic pulses using the first ultrasonic transmitter; at a seventh time, receive the third set of ultrasonic pulses using the second ultrasonic receiver; at an eighth time, transmit a fourth set of ultrasonic pulses using the second ultrasonic transmitter; at a ninth time,

receive a first echo of multiple echoes of the third set of ultrasonic pulses using the second ultrasonic receiver, in which the eighth time is between the seventh time and the ninth time; at a tenth time, receive the fourth set of ultrasonic pulses using the first ultrasonic receiver; and provide an update to the flow rate measurement between the first and second locations based on the sixth time, the seventh time, the eighth time, and the tenth time.

4. The apparatus of claim 3, wherein the one or more controllers are configured to: transmit the first set of ultrasound pulses and the second set of ultrasound pulses are at a first pulse frequency; transmit the third set of ultrasound pulses and the fourth set of ultrasound pulses are at a second pulse frequency, wherein the first pulse frequency and the second pulse frequency are different.

5. The apparatus of claim 4, wherein the one or more controllers are configured to set the first pulse frequency and the second pulse frequency based on frequency responses of the first ultrasonic transmitter, the first ultrasonic receiver, the second ultrasonic transmitter, and the second ultrasonic receiver, wherein the frequency responses of the first ultrasonic transmitter, the first ultrasonic receiver, the second ultrasonic transmitter, and the second ultrasonic receiver have multiple resonance peaks.

6. The apparatus of claim 3, wherein the one or more controllers are configured to: transmit the first set of ultrasound pulses and the second set of ultrasound pulses as upchirp pulse signals having a first pulse frequency range; transmit the third set of ultrasound pulses and the fourth set of ultrasound pulses as downchirp pulse signals having a second pulse frequency range.

7. The apparatus of claim 6, wherein: the first pulse frequency range and the second pulse frequency range overlap.

8. The apparatus of claim 6, wherein: the first pulse frequency range and the second pulse frequency range do not overlap.

9. An apparatus, comprising: a first ultrasonic transmitter; a first ultrasonic receiver; and a controller coupled to the first ultrasonic transmitter and the first ultrasonic receiver, the controller having a controller input and a measurement output and configured to: receive a first signal at the controller input, the first signal representing a first time of transmission of a first set of ultrasonic pulses by a second ultrasonic transmitter; at a second time, receive the first set of ultrasonic pulses using the first ultrasonic receiver; based on the first time and the second time, determine an expected arrival time of a first echo of multiple echoes of the first set of ultrasonic pulses at the first ultrasonic receiver; determine a third time of transmission of a second set of ultrasonic pulses, in which the third time precedes the expected arrival time; at the third time, transmit a second set of ultrasonic pulses using the first ultrasonic transmitter; and provide a second signal representing a flow rate measurement at the measurement output based on the first, second, and third times.

10. The apparatus of claim 9, wherein: the first ultrasonic transmitter, the first ultrasonic receiver, and the controller are included in a single chip.

11. The apparatus of claim 9, wherein: the first ultrasonic transmitter and the first ultrasonic receiver are at a first location, and the second ultrasonic transmitter and a second ultrasonic receiver are at a second location.

12. The apparatus of claim 11, wherein the controller is configured to: receive a third signal at the controller input, the third signal including data representing a distance between the first and second locations and an angle of propagation of the first and second sets of ultrasonic pulses between the first and second locations; receive a fourth signal at the controller input, the fourth signal representing a fifth time of receiving of the second set of ultrasonic pulses by the second ultrasonic receiver; determine a first time of flight (TOF) of the first set of ultrasonic pulses propagating from the second location to the first location based on the first time and the second time; determine a second TOF of the second set of ultrasonic pulses propagating from the first location to the second location based on the third time and the fifth time; and provide a fifth signal representing an update to the flow rate measurement at the measurement output based on the first TOF, the second TOF, and the data.

13. A method, comprising: at a first time, transmitting a first set of ultrasonic pulses from a first location; at a second time, receiving the first set of ultrasonic pulses at a second location; at a third time, transmitting a second set of ultrasonic pulses from the second location; at a fourth time, receiving a first echo of multiple echoes of the first set of ultrasonic pulses at the second location, in which the third time is the same as the first time, or is between the second time and the fourth time; at a fifth time, receiving the second set of ultrasonic pulses at the first location; and providing a flow rate measurement between the first and second locations based on the first time, the second time, the third time, and the fifth time.

14. The method of claim 13, further comprising: at a sixth time, transmitting a third set of ultrasonic pulses from the first location; at a seventh time, receiving the third set of ultrasonic pulses at the second location; at an eighth time, transmitting a fourth set of ultrasonic pulses from the second location; at a ninth time, receiving a first echo of multiple echoes of the third set of ultrasonic pulses at the second location, in which the eighth time is between the seventh time and the ninth time; at a tenth time, receiving the fourth set of ultrasonic pulses at the first location; and providing an update to the flow rate measurement between the first and second locations based on the sixth time, the seventh time, the eighth time, and the tenth time.

15. The method of claim 14, further comprising: transmitting the first set of ultrasound pulses and the second set of ultrasound pulses are at a first pulse frequency; and transmitting the third set of ultrasound pulses and the fourth set of ultrasound pulses are at a second pulse frequency, wherein the first pulse frequency and the second pulse frequency are different.

16. The method of claim 15, further comprising: setting the first pulse frequency and the second pulse frequency based on frequency responses of one or more ultrasonic transmitters and ultrasonic receivers, wherein the frequency responses of the one or more ultrasonic transmitters and ultrasonic receivers have multiple resonance peaks.

17. The method of claim 14, further comprising: transmitting the first set of ultrasound pulses and the second set of ultrasound pulses as upchirp pulse signals having a first pulse frequency range; and transmitting the third set of ultrasound pulses and the fourth set of ultrasound pulses as downchirp pulse signals having a second pulse frequency range.

18. The method of claim 17, wherein: the first pulse frequency range and the second pulse frequency range overlap.

19. The method of claim 17, wherein: the first pulse frequency range and the second pulse frequency range do not overlap.

20. A non-transitory computer readable medium storing instructions that, when executed by processor circuitry, cause the processor circuitry to control a first ultrasonic transmitter and a first ultrasonic receiver at a first location, a second ultrasonic transmitter and a second ultrasonic receiver at a second location to: at a first time, transmit a first set of ultrasonic pulses from a first location; at a second time, receive the first set of ultrasonic pulses at a second location; at a third time, transmit a second set of ultrasonic pulses from the second location; at a fourth time, receive a first echo of multiple echoes of the first set of ultrasonic pulses at the second location, in which the third time is the same as the first time, or is between the second time and the fourth time; at a fifth time, receive the second set of ultrasonic pulses at the first location; and provide a flow rate measurement between the first and second locations based on the first time, the second time, the third time, and the fifth time.
